BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Integrated Development Environments

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does an interpreter normally do?

2. Q = (A AND B) OR C. A=1, B=0, C=1. What is Q?
   A) 0
   B) 2
   C) 1
   D) 10

3. "A count-controlled loop is only suitable when the number of repetitions can never be known." Correct this. Explain why.

4. A program keeps asking for a password until its stopping condition is met. "A condition-controlled loop always runs ten times." Correct this. What controls when it stops?

Today's objectives

Explain the core ideas of integrated development environments using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Editor

Provides tools for entering and organising source code.

Syntax highlighting and completion

Make code structure and possible mistakes easier to notice.

Breakpoints and debugging

Pause execution to inspect variables and follow faulty logic.

Runtime/output tools

Allow programmers to run/test code and view results from the environment.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Editor
Provides tools for entering and organising source code.

Syntax highlighting and completion
Make code structure and possible mistakes easier to notice.

Breakpoints and debugging
Pause execution to inspect variables and follow faulty logic.

Runtime/output tools
Allow programmers to run/test code and view results from the environment.

I Do - Worked model

A program produces 9 instead of expected 10. Explain how a debugger could locate the error.

- Set a breakpoint before the suspect calculation.
- Run to the breakpoint.
- Inspect current variable values and step line by line.
- Find first unexpected update, fix it and rerun tests.

We Do - Guided practice

1. A pupil has misspelled a variable. Identify two IDE features that may help.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the faulty bits-to-bytes calculation and check its output.

```
bits = 40
bytes = bits * 8
print(bytes)
```

You Do - Independent exam questions

Q2. Name two IDE features. [2 marks]

Q3. Explain why a breakpoint helps a developer. [3 marks]

Q4. State why a program should be tested after fixing an IDE-reported syntax error. [2 marks]

Exit Ticket - True or False?

A breakpoint changes all incorrect variables automatically.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. An interpreter translates and runs code as it goes, often one statement at a time. This gives quick feedback during testing.

2. C - 1

3. When the number of repetitions is known or determined by a fixed count.

4. A condition-controlled loop repeats until a condition tells it to stop. It need not run ten times. Choose the loop to suit the task and trace its steps.

I Do - Explain the model

Set a breakpoint before the suspect calculation.

Run to the breakpoint.

Inspect current variable values and step line by line.

Find first unexpected update, fix it and rerun tests.

We Do - Model answers

1. Test after correction.

Code First - Answer

```
bits = 40
bytes = bits DIV 8
print(bytes)
```

Expected output: 5

You Do - Self-mark

Q2: Code editor, syntax highlighting, autocomplete, debugging, breakpoints, error diagnostics or run console.

Q3: Pauses execution at a selected point, allowing variables/program state to be inspected.

Q4: Syntax correction only ensures code can be parsed; logical behaviour still needs checking.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.